

Sara Robin Wilson

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Education

Michigan State University Dual Ph.D. in Computational Mathematics, Science, and Engineering & Physics <i>Advised by Prof. Andrew Christlieb and Prof. Brian O'Shea</i>	East Lansing, MI Expected May 2031
University of Pittsburgh B.S. in Mathematics	Pittsburgh, PA May 2026

Research Interests

computational plasma physics; numerical analysis of PDEs; scientific machine learning

Honors & Awards

Edna M. Heck Scholarship
Society of Women Engineers Certificate of Merit in Math and Science

Publications

An Analysis of a 2×2 Keyfitz-Kranzer Type Balance System with Varying Generalized Chaplygin Gas <i>Physics of Fluids</i> J. Frew, N. Keyser, E. Kim, G. Paddock, C. Tumbleston, S. Wilson , C. Tsikkou 10.1063/5.0231413 	Sept 2024
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Research Experience

Argonne National Laboratory Internship Research on numerical stability of hyperbolic PDE solvers under reduced precision.	Lemont, IL Jun 2026 - Aug 2026
Carnegie Mellon University DRP Study of computational plasma physics and discontinuous Galerkin methods.	Pittsburgh, PA Sept 2025 – Apr 2026
Semiconductor Research Corporation REU Research on design modifications for wireless power transfer.	Tuscaloosa, AL Jun 2025 – Jul 2025
West Virginia University REU Research on numerical analysis of 2×2 Keyfitz-Kranzer type balance system with varying Chaplygin gas.	Morgantown, WV Jun 2024 – Jul 2024

Skills

Programming: Python, MATLAB, Java
Scientific Computing: FEniCS, ParaView
Tools: Git, LaTeX

Talks & Presentations

CMU–Pitt Directed Reading Program <i>Nodal Discontinuous Galerkin Methods</i>	Pittsburgh, PA Apr 2026
CMU–Pitt Directed Reading Program <i>Computational Plasma Physics</i>	Pittsburgh, PA Dec 2025
Summer Research Symposium, University of Alabama <i>Wireless Power Transfer</i>	Tuscaloosa, AL Jul 2025
Pi Mu Epsilon Poster Session, Joint Mathematics Meetings <i>Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System</i>	Seattle, WA Jan 2025

Summer Research Symposium, West Virginia University
Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System

Morgantown, WV
Jul 2024

Additional Training

Summer School in Fluids and Computer Assisted Proofs
Princeton University

Princeton, NJ
Aug 2025

Summer School in Modeling and Simulation of PDEs
Texas A&M University

College Station, TX
May 2025

Teaching Experience

University of Pittsburgh
Teaching Assistant

Pittsburgh, PA
Aug 2024 – Apr 2026

University of Pittsburgh
Mathematics Tutor

Pittsburgh, PA
Jan 2023 – Apr 2026

West Virginia University
REU Mentor

Remote
Jun 2025 - Jul 2025